An Analysis of GPT language model

GPT language model

GPT language model -- Part 1

Background During the 2010s, improved machine learning algorithms, more powerful computers, and an increase in the amount of digitized material allowed for an AI boom. Separately, the concept of generative pre-training (GP) was a long-established technique in machine learning. GP is a form of self-supervised learning wherein a model is first trained on a large, unlabeled dataset (the "pre-training" step) to learn to generate data points. This pre-trained model is then adapted to a specific task using a labeled dataset (the "fine-tuning" step). The transformer architecture for deep learning is the core technology of a GPT. Developed by researchers at Google, it was introduced in the paper "Attention Is All You Need", which was released on June 12, 2017. The transformer architecture solved many of the performance issues that were associated with older recurrent neural network (RNN) designs for natural language processing (NLP). The architecture's use of an attention mechanism allows models to process entire sequences of text at once, enabling the training of much larger and more sophisticated models. Since 2017, available transformer-based NLP systems have been capable of processing, mining, organizing, connecting, contrasting, and summarizing texts as well as answering questions from textual input.

GPT language model -- Part 2

Evaluation and benchmarking The evaluation of generative pre-trained transformer models is carried out using a range of benchmarks and metrics, which aim to assess their performance across different tasks. Common approaches include accuracy on standard datasets, as well as other features such as robustness, bias, and toxicity. These models are typically tested on tasks such as natural language understanding, reasoning, answering queries, and code generation. Sometimes they combine multiple tasks to provide a broader assessment of model performance across domains. More recent approaches extend these evaluations to include other features such as fairness, efficiency, and transparency, in order to get a more accurate assessment of these models. Evaluation remains an active area of research, as existing tests may not accurately reflect real-world performance or the risks associated with large-scale generative models.

GPT language model -- Part 3

OpenAI, which created the first generative pre-trained transformer (GPT) in 2018, asserted in 2023 that "GPT" should be regarded as a brand of OpenAI. In April 2023, OpenAI revised the brand guidelines in its terms of service to indicate that other businesses using its API to run their AI services would no longer be able to include "GPT" in such names or branding. In May 2023, OpenAI engaged a brand management service to notify its API customers of this policy, although these notifications stopped short of making overt legal claims (such as allegations of trademark infringement or demands to cease and desist). As of November 2023, OpenAI still prohibits its API licensees from naming their own products with "GPT", but it has begun enabling its ChatGPT Plus subscribers to make "custom versions of ChatGPT" called GPTs on the OpenAI site. OpenAI's terms of service says that its subscribers may use "GPT" in the names of these, although it's "discouraged". Relatedly, OpenAI has applied to the United States Patent and Trademark Office (USPTO) to seek domestic trademark registration for the term "GPT" in the field of AI. OpenAI sought to expedite handling of its application, but the USPTO declined that request in April 2023. In May 2023, the USPTO responded to the application with a determination that "GPT" was both descriptive and generic. As of November 2023, OpenAI continued to pursue its argument. For any given type or scope of trademark protection in the U.S., OpenAI would need to establish that the term is actually "distinctive" to their specific offerings in addition to being a broader technical term for the kind of technology. Some media reports suggested in 2023 that OpenAI may be able to obtain trademark registration based indirectly on the fame of its GPT-based chatbot product, ChatGPT, for which OpenAI has separately sought protection (and which it has sought to enforce more strongly). Other reports have indicated that registration for the bare term "GPT" seems unlikely to be granted, as it is used frequently as a common term to refer simply to AI systems that involve generative pre-trained transformers. In any event, to whatever extent exclusive rights in the term may occur the U.S., others would need to avoid using it for similar products or services in ways likely to cause confusion. If such rights ever became broad enough to implicate other well-established uses in the field, the trademark doctrine of descriptive fair use could still continue non-brand-related usage. In the European Union, the European Union Intellectual Property Office registered "GPT" as a trade mark of OpenAI in spring 2023. However, since spring 2024 the registration is being challenged and is pending cancellation. In Switzerland, the Swiss Federal Institute of Intellectual Property registered "GPT" as a trade mark of OpenAI in spring 2023.

GPT language model -- Part 4

Emergent abilities Emergent abilities refer to capabilities that appear in large language models only when they reach a certain scale and are not present in smaller versions of the same models. These abilities are considered "emergent" because they arise as model size, training data, and compute increase. Examples of emergent abilities include multi-step reasoning, in-context learning (the ability to perform tasks based on examples provided in prompts without additional training), and improved performance on complex language and reasoning benchmarks. Research suggests that these capabilities do not scale linearly, but instead appear once models exceed certain thresholds in size and training scale. This phenomenon has influenced the development of larger GPT models and contributed to their increased effectiveness across a wide range of tasks.

GPT language model -- Part 5

EinsteinGPT – for sales and marketing domains, to aid with customer relationship management (uses GPT-3.5) BloombergGPT – for the financial domain, to aid with financial news and information (uses "freely available" AI methods, combined with their proprietary data) Khanmigo – described as a GPT version for tutoring, in the education domain, it aids students using Khan Academy by guiding them through their studies without directly providing answers (powered by GPT-4) SlackGPT – for the Slack instant-messaging service, to aid with navigating and summarizing discussions on it (uses OpenAI's API) BioGPT – for the biomedical domain, to aid with biomedical literature text generation and mining (uses GPT-2) Sometimes domain-specificity is accomplished via software plug-ins or add-ons. For example, several different companies have developed particular plugins that interact directly with OpenAI's ChatGPT interface, and Google Workspace has available add-ons such as "GPT for Sheets and Docs" – which is reported to aid use of spreadsheet functionality in Google Sheets.